BMC Public Health

RESEARCH

Open Access

Co-exposure to heat and noise on workers' health: evidence from a large-scale cross-sectional surveillance study in China

Minhua Li[1,2†], Yilin Zhang[1,2†], Chuancheng Wu[1,2], Yu Jiang[1,2], Li Jing[1,2], Jing Wu[1,2], Jinfeng Lin[3*] and Jianjun Xiang[1,2*]

Abstract

Objectives This study aimed to investigate the synergistic effects of co-exposure to occupational heat and noise on multi-system health outcomes among Chinese workers, and to examine the mediating role of systemic inflammation, indicated by white blood cell (WBC) count.

Methods A cross-sectional analysis was conducted using data from the Fujian Workplace Occupational Hazards Comprehensive Surveillance Program (2020–2022), encompassing 10,275 workers from the manufacturing, petrochemical, mining, and construction industries. Multivariate logistic regression, relative excess risk due to interaction (RERI), random forest with SHAP values, and mediation analysis were used to explore interaction effects and mediating pathways.

Results Co-exposure to heat and noise in the workplace was associated with increased risks of hypertension (OR = 1.94, 95%CI:1.61–2.36; RERI = 0.26, *P* interaction <0.001) and abnormal ECG (OR = 1.81, 95%CI:1.48–2.12; RERI = 0.16, *P* interaction <0.001), indicating synergistic effects beyond individual exposures. WBC partially mediated the effects of occupational heat and noise exposure on hypertension (heat: 9.2%; noise: 7.4%; co-exposure: 15.3%), liver dysfunction (heat: 5.4%; noise: 13.2%; co-exposure: 11.5%), and abnormal ECG findings (heat: 7.0%; noise: 6.4%; co-exposure: 8.7%). Machine learning models further confirmed that interaction between heat and noise was a stronger predictor of adverse health outcomes than either exposure alone.

Conclusion Co-exposure to occupational heat and noise significantly exacerbates health risks across multiple organ systems, with systemic inflammation playing a partial mediating role. These findings highlight the importance of integrated occupational hazard assessments and targeted interventions that consider multiple concurrent occupational hazards.

Keywords Occupational health, Heat exposure, Noise exposure, Co-exposure, Interaction

†Yilin Zhang and Minhua Li contributed equally to this work.

*Correspondence:
Jinfeng Lin
ljf9804002@163.com
Jianjun Xiang
jianjun.xiang@fjmu.edu.cn

[1]Department of Preventive Medicine, School of Public Health, Fujian Medical University, 1 North Xue-Fu Road, Fuzhou 350122, Fujian Province, China
[2]Key Laboratory of Environment and Health of Fujian Higher Education Institutes, School of Public Health, Fujian Medical University, Fuzhou 350122, Fujian Province, China
[3]Fujian Center for Prevention and Control of Occupational Diseases and Chemical Poisoning, Fuzhou 350025, Fujian Province, China

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

Introduction

With the intensification of ongoing climate change, the risk of heat exposure in occupational environments has become increasingly prominent, while the continuous advancement of industrialization has also made noise pollution an occupational health threat that cannot be ignored. As the most representative physical hazards in industrial settings, occupational heat and noise exposure have been proven to be closely associated with adverse effects on workers' health, safety, and labor productivity [1–4]. A systematic review encompassing multiple industries, including agriculture, construction, mining, and manufacturing, demonstrated that extreme heat exposure can negatively impact health outcomes such as the cardiovascular, nervous, and respiratory systems [5]. Similarly, a systematic review and meta-analysis covering 30 countries and 447 million workers also confirmed the impact of occupational heat exposure on workers' health [6]. Likewise, the health effects of occupational noise exposure have been well documented. A UK-based prospective cohort study revealed a significant association between workplace noise exposure and the incidence of atrial fibrillation [7]. In addition, the WHO/ILO Joint Estimates of the Work-related Burden of Disease and Injury also identified occupational noise exposure as a contributing factor to ischemic heart disease, stroke, and hypertension [8]. However, it is noteworthy that in real-world industrial environments, heat and noise exposures often occur simultaneously. Most existing studies focus on the effects of these hazards in isolation or examine single health outcomes. The potential synergistic effects of combined heat and noise exposure on multiple body systems remain underexplored, creating a critical gap in our understanding of complex occupational risk profiles [9, 10].

White blood cell (WBC) count, a sensitive biomarker of systemic inflammation, has emerged as a potential mediator in the relationship between environmental stressors and adverse health outcomes [11–13]. Elevated WBC counts have been linked to various occupational exposures and may serve as an early predictor of inflammatory responses. For instance, foundry workers in Iran exhibited higher WBC levels in association with elevated Wet Bulb Globe Temperature (WBGT) index values [14]. A cohort study showed that immune parameters, including WBC count, can reflect the physiological impact of heat exposure [15]. A literature review incorporating studies from multiple countries revealed that firefighters exposed to extreme heat exhibited significantly elevated hematological parameters and inflammatory markers [16]. Similarly, a case-control study conducted in an insulation material factory in Iran showed that noise-exposed production workers had significantly higher WBC counts compared to administrative staff [17]. These findings suggest that WBC count may play an important role in the biological pathways linking occupational heat or noise exposure to health impairments. However, most existing evidence comes from studies of single-exposure settings, and few large-scale investigations have examined the mediating role of WBCs in multiple health outcomes under combined heat and noise exposures in the workplace.

To address this research gap, we analyzed the data obtained from the Fujian Workplace Occupational Hazards Comprehensive Surveillance Program in China. We assessed the individual and joint effects of occupational heat and noise exposure on cardiovascular, respiratory, hepatic, and renal functions, using relative excess risk due to interaction (RERI) and machine learning models. Furthermore, we conducted mediation analysis to explore the potential mediating role of systemic inflammation, as indicated by WBC count, in these associations. Our findings will provide novel insights into the health effects of co-exposure to heat and noise in the workplace and inform integrated strategies for occupational hazard management.

Methods

Study design and participants

This cross-sectional study was conducted in accordance with the "2021 Occupational Disease Prevention and Control Implementation Plan" issued by the General Office of China's National Health Commission. Data for this analysis were derived from the Fujian Workplace Occupational Hazards Comprehensive Surveillance Program, which collected occupational health monitoring data from workers across multiple industries in Fujian Province between January 1, 2020, and December 31, 2022. Coordinated by the Fujian Center for Prevention and Control of Occupational Diseases and Chemical Poisoning, monitoring activities covered all prefecture-level cities and the Pingtan Comprehensive Experimental Zone of Fujian Province. Data collection encompassed various sectors, including the service sector, petrochemical, mining, cement, construction, and other manufacturing industries such as metal processing and thermal power generation. The dataset included demographic and occupational information such as age, gender, work duration, income level, company size, protective measures (e.g., cooling vests, earplugs, helmets, gloves, etc.), occupational exposure profiles, as well as workers' health monitoring results, including routine biochemical tests, hypertension monitoring, blood glucose tests, electrocardiograms, pulmonary function parameters, liver function indices, and renal function markers.

Participants meeting the following criteria were included in the analysis: workers who participated in the Fujian Workplace Occupational Hazards Comprehensive

Surveillance Program between January 1, 2020, and December 31, 2022, with complete occupational records and at least one year of work experience in industries. Exclusion criteria consisted of individuals with pre-existing severe cardiovascular diseases, chronic respiratory diseases, liver failure, chronic kidney disease, malignancies, mental disorders, or physical disabilities at the time of pre-employment health screening.

Heat exposure assessment and definition

Occupational heat exposure assessment employs portable WBGT monitors (3 M QUESTemp°34) for on-site measurements using the WBGT index as the core metric. The measurement procedures adhere to the technical requirements specified in GBZ/T 224–2010 "Technical Specifications for Occupational Hazard Monitoring," with monitoring point arrangements considering working posture and heat exposure uniformity. For standing postures, the baseline measurement height is 1.5 m (respiratory zone height), while for sitting postures, it is 1.1 m. Monitoring points comprehensively cover key areas, including core production zones, auxiliary work areas, and boundary locations. In the absence of production-related heat sources, three monitoring points are established, whereas 3 to 5 points are set when heat sources are present, with the average value of all points taken as the final result. For mobile workers, relatively fixed working points within their activity range are selected for measurement, and the time-weighted WBGT index is calculated.

Regarding measurement timing, when the heat source in the work environment is stable, measurements are conducted three times daily: 0.5 h after work begins, 0.5 h before work ends, and once during the middle of the work shift, with the average value taken as the final result. If the heat sources in the work environment are unstable and the production process undergoes significant periodic variations, the time-weighted average WBGT index is measured and calculated separately. Data collection spanned five consecutive typical workdays to ensure coverage of the operational cycle.

Workers were categorized as "exposed" to heat if the measured WBGT values were ≥ 25 °C, in line with relevant national standards [18].

Noise exposure assessment and definition

Occupational noise exposure assessment was conducted in accordance with the Chinese National Standard GBZ/T 229.4–2012 "Measurement of Physical Agents in the Workplace – Part 4: Noise." A preliminary rapid scan was conducted to identify peak noise areas across different work environments. Fixed-point measurements were then performed at ear height (approximately 1.5 m) at representative locations using a Cirrus ASV5910 Personal Sound Exposure Meter. At each point, steady-state and non-steady-state noise were measured using A-weighting, while impulse noise was measured using C-weighting. For steady-state and non-steady-state noise, the equivalent continuous A-weighted sound pressure level (LAeq, T) was recorded, and for impulse noise, the peak C-weighted sound pressure level (LC, Peak) was obtained. Monitoring was carried out over five consecutive typical workdays to ensure coverage of daily noise variations.

Based on national occupational health criteria [19], workers were classified as noise-exposed if their 8-hour equivalent continuous A-weighted sound pressure level (LEX, 8h) was ≥ 80 dB(A).

Health outcomes

Hypertension

Blood pressure was measured using an OMRON electronic sphygmomanometer (Model: HEM-7121), following the protocol recommended by the American Heart Association [20–22]. The specific protocol included:

(1) Participants refrained from vigorous physical activity, caffeine intake, and smoking for at least 30 min prior to the measurement.
(2) Participants were required to rest in a seated position for at least 5 min before measurement. During measurement, they remained seated with their back straight, feet flat on the floor, and arm supported at heart level.
(3) The measurements were repeated 3 times with a1-minute interval and the average of 3 readings was used for analysis.
(4) During the first measurement, blood pressure was recorded in both arms, and subsequent measurements were taken on the arm with the higher reading. If the difference between the two arms was ≤ 10 mmHg, the right arm was used for all subsequent measurements.

Hypertension was diagnosed according to the WHO/ISH criteria: systolic blood pressure ≥ 140 mmHg or diastolic blood pressure ≥ 90 mmHg [23].

Diabetes

Diabetes can be diagnosed if any one of the following criteria is met [24]:

(1) Fasting plasma glucose (FPG) ≥ 7.0 mmol/L (126 mg/dL).
(2) 2-hour plasma glucose during an oral glucose tolerance test (OGTT) ≥ 11.1 mmol/L (200 mg/dL).
(3) Glycated hemoglobin (HbA1c) ≥ 6.5% (48 mmol/mol).

(4) Random plasma glucose ≥ 11.1 mmol/L (200 mg/dL).

Renal dysfunction

Renal dysfunction can be diagnosed if either of the following criteria is met [25, 26]:

Estimated glomerular filtration rate (eGFR) < 60 mL/min/1.73 m^2.

Urinary albumin-to-creatinine ratio (UACR) ≥ 30 mg/g.

Pulmonary dysfunction

Pulmonary function tests, including forced vital capacity (FVC), forced expiratory volume in 1 s (FEV_1), and the FEV_1/FVC ratio, were performed by trained technicians following standardized protocols. Abnormal pulmonary function tests was defined according to the American Thoracic Society/European Respiratory Society (ATS/ERS) guidelines based on the following criteria for ventilatory impairment: for an obstructive ventilatory defect, FEV_1/FVC < 0.7 (fixed ratio) or < lower limit of normal (LLN) plus FEV_1 < 80% predicted; for a restrictive ventilatory defect, FVC < 80% predicted plus FEV_1/FVC ≥ LLN plus total lung capacity (TLC) < 80% predicted; and for a mixed ventilatory defect, FEV_1/FVC < LLN plus FVC < 80% predicted [27, 28]. In this study, pulmonary dysfunction was defined as meeting any one of the above criteria.

Liver dysfunction

Liver dysfunction was defined as any of the following biomarkers exceeding the normal range: ALT, AST, ALP, GGT, or TBIL. The reference values for these biomarkers are as follows: AST: 0–37 U/L in men and 0–31 U/L in women; ALT: 0–40 U/L for males, 0–31 U/L for females; ALP: Male 40–129 U/L, female 35–104 U/L; GGT: Male 11–51 U/L, female 7–33 U/L; TBIL: 1.0 mg/dL(17.1μmol/L) [29].

Abnormal ECG

Abnormal electrocardiograms (ECGs) were defined as any findings other than normal sinus rhythm and were coded according to the Minnesota Code (MC) [30, 31]. These included, but were not limited to, the following MC classifications: Q-QS wave abnormalities (MC 1–1 to 1-2-8), left ventricular hypertrophy (MC 3 – 1), Wolff-Parkinson-White (WPW) syndrome (MC 6-4-1 or 6-4-2), complete bundle branch block or intraventricular conduction delay (MC 7-1-1, 7-2-1, 7 – 4, or 7–8), atrial fibrillation or atrial flutter (MC 8 – 3), major ST-T segment abnormalities (MC 4 – 1, 4 – 2, 5 – 1, or 5 – 2), and minor ST-T segment changes (MC 4 – 3, 4–4, 5 – 3, or 5 – 4). All ECGs were recorded using standard 12-lead configurations (leads I, II, III, aVR, aVL, aVF, and V1–V6), with participants resting in the supine position for at least 20 min prior to measurement to ensure data accuracy. Each ECG tracing was independently interpreted by two cardiologists to guarantee consistency and reliability in the assessments.

WBC count measurement

Blood samples were collected in the morning following an 8–12-hour overnight fast. Venous blood was drawn into EDTA-anticoagulated tubes and analyzed within 2 h using an automated hematology analyzer (e.g., Mindray BS-2000 M) to ensure sample integrity. All testing laboratories involved in this study were accredited by relevant authorities such as the National Center for Clinical Laboratories (NCCL) of China and operated under standardized protocols with routine internal and external quality control procedures to ensure measurement accuracy and consistency across sites.

Statistical analysis

Descriptive statistics were used to summarize the characteristics of study participants by exposure group. Continuous variables were expressed as means (± SD) or medians (IQR), and categorical variables as frequencies and percentages. Group comparisons were performed using the chi-square tests or Fisher's exact tests for categorical variables, and independent t-tests or Mann–Whitney U tests for continuous variables for two-group comparisons; one-way ANOVA was used for multi-group comparison.

Multivariable logistic regression models were used to estimate the independent and joint associations of occupational heat and noise with various health outcomes, reporting OR with 95% confidence intervals (CI). The relative excess risk due to interaction (RERI) was calculated to assess potential synergistic effects between occupational exposures. To capture complex interactions that might be overlooked by conventional regression models, we employed a random forest machine learning algorithm. Predictive contributions of variables were further analyzed using SHAP values via the fastshap package. Finally, a counterfactual framework was used for mediation analysis to quantify the proportion of the effect of heat and noise exposure on health outcomes mediated by WBC count.

All statistical analyses were conducted using R (version 4.2.2) and Python (version 3.9). A two-sided p-value < 0.05 was considered statistically significant.

Ethics approval and consent to participate

This study and its protocol were ethically approved by Fujian Medical University's Biomedical Research Ethics Committee (Approval No. 2022-111). The monitoring activities were conducted across all 9 prefecture-level cities in Fujian Province and the Pingtan Comprehensive Experimental Zone, covering all subordinate counties

and districts. The Fujian Provincial Center for Occupational Disease and Chemical Poisoning Prevention and Control was responsible for assigning workplace occupational hazard monitoring tasks across the province, with local governments clearly communicating these to relevant manufacturing entities. The leading department of each manufacturing enterprise was tasked with obtaining informed written consent from all participants. Participation in health monitoring was voluntary, and individuals retained the right to refuse or withdraw from the survey at any time based on their personal circumstances. Throughout the data collection process and the entire duration of the investigation, all participant information was kept strictly confidential. Furthermore, all methods employed in this study strictly adhere to the principles outlined in the Declaration of Helsinki.

Results

Characteristics of study participants

This study assessed the health effects of occupational heat and noise exposure among 10,275 workers. As shown in Table 1, compared to the non-heat-exposed group (N=6,233), workers in the heat-exposed group (N=4,042) were significantly older (41.29 ± 9.70 vs. 40.34 ± 10.31, $p < 0.001$), had a higher proportion of males (86.2% vs. 82.9%, $p < 0.001$), and longer employment duration (13.44 ± 9.87 vs. 11.12 ± 9.80 years, $p < 0.001$). Regarding health status, workers exposed to heat in the workplace exhibited higher prevalence rates of hypertension (35.1% vs. 24.2%), renal dysfunction (20.6% vs. 16.5%), liver dysfunction (31.2% vs. 25.4%), pulmonary dysfunction (26.9% vs. 23.6%), and abnormal ECG findings (32.2% vs. 26.3%) (all $p < 0.05$) compared to those

Table 1 Participant characteristics by heat or noise exposure status

Variables	Overall	Non-heat	Heat	*P*-value	Non-noise	Noise	*P*-value
	N = 10,275	*N* = 6233	*N* = 4042		*N* = 6437	*N* = 3838	
Age(years)	40.71 ± 10.08	40.34 ± 10.31	41.29 ± 9.70	< 0.001	40.52 ± 10.21	41.03 ± 9.86	0.013
Employment duration (years)	12.03 ± 9.89	11.12 ± 9.80	13.44 ± 9.87	< 0.001	11.87 ± 10.14	12.30 ± 9.46	0.033
Gender, n (%)				< 0.001			0.005
Male	8,652 (84.2%)	5,166 (82.9%)	3,486 (86.2%)		5,370 (83.4%)	3,282 (85.5%)	
Female	1,623 (15.8%)	1,067 (17.1%)	556 (13.8%)		1,067 (16.6%)	556 (14.5%)	
Income level, n (%)				< 0.001			0.018
Low	8,696 (84.6%)	5,308 (85.2%)	3,388 (83.8%)		5,454 (84.7%)	3,242 (84.5%)	
Medium	1,231 (12.0%)	679 (10.9%)	552 (13.7%)		744 (11.6%)	487 (12.7%)	
High	348 (3.4%)	246 (3.9%)	102 (2.5%)		239 (3.7%)	109 (2.8%)	
Company size, n (%)				0.613			0.015
Small	4,592 (45.4%)	2,798 (45.6%)	1,794 (45.0%)		2,827 (44.4%)	1,765 (46.9%)	
Large	5,533 (54.6%)	3,343 (54.4%)	2,190 (55.0%)		3,537 (55.6%)	1,996 (53.1%)	
Protect measure, n (%)				< 0.001			0.720
No	4,225 (41.1%)	2,318 (37.2%)	1,907 (47.2%)		2,656 (41.3%)	1,569 (40.9%)	
Yes	6,050 (58.9%)	3,915 (62.8%)	2,135 (52.8%)		3,781 (58.7%)	2,269 (59.1%)	
Hypertension, n (%)				< 0.001			< 0.001
No	7,347 (71.5%)	4,724 (75.8%)	2,623 (64.9%)		4,763 (74.0%)	2,584 (67.3%)	
Yes	2,928 (28.5%)	1,509 (24.2%)	1,419 (35.1%)		1,674 (26.0%)	1,254 (32.7%)	
Diabetes, n (%)				0.313			0.128
No	7,904 (92.7%)	4,341 (93.0%)	3,563 (92.4%)		4,979 (93.1%)	2,925 (92.2%)	
Yes	620 (7.3%)	327 (7.0%)	293 (7.6%)		371 (6.9%)	249 (7.8%)	
Renal dysfunction, n (%)				< 0.001			0.448
No	6,826 (81.8%)	4,115 (83.5%)	2,711 (79.4%)		4,294 (82.1%)	2,532 (81.4%)	
Yes	1,514 (18.2%)	812 (16.5%)	702 (20.6%)		936 (17.9%)	578 (18.6%)	
Liver dysfunction, n (%)				< 0.001			0.237
No	6,198 (72.1%)	3,590 (74.6%)	2,608 (68.8%)		3,952 (72.5%)	2,246 (71.3%)	
Yes	2,402 (27.9%)	1,222 (25.4%)	1,180 (31.2%)		1,498 (27.5%)	904 (28.7%)	
Pulmonary dysfunction, n (%)				0.004			0.035
No	4,993 (75.2%)	3,297 (76.4%)	1,696 (73.1%)		3,078 (76.2%)	1,915 (73.8%)	
Yes	1,643 (24.8%)	1,020 (23.6%)	623 (26.9%)		964 (23.8%)	679 (26.2%)	
Abnormal ECG, n (%)				< 0.001			< 0.001
No	7,297 (71.0%)	4,595 (73.7%)	2,702 (66.8%)		4,778 (74.2%)	2,519 (65.6%)	
Yes	2,978 (29.0%)	1,638 (26.3%)	1,340 (33.2%)		1,659 (25.8%)	1,319 (34.4%)	

Note: Table 1 presents the characteristics of workers grouped by their exposure status to either heat or noise. Specifically, the heat-exposed group may include participants with or without noise exposure, and similarly, the noise-exposed group may include participants with or without heat exposure

without heat exposure. Similar patterns were observed among those exposed to noise, who were older, more likely to be male, had longer employment duration, and significantly increased prevalence of hypertension, pulmonary dysfunction, and abnormal ECG compared to those without noise exposure.

Table 2 further categorized participants into four groups based on co-exposure status: no-exposure group (N=4,330), heat-only group (N=2,107), noise-only group (N=1,903), and dual-exposure group (N=1,935). Results demonstrated that the dual-exposure group had the worst health outcomes, with significantly higher prevalence rates of hypertension (38.2%), liver dysfunction (32.3%), pulmonary dysfunction (28.1%), and abnormal ECG findings (37.8%) among the four groups. Additionally, both the heat-only and noise-only groups exhibited significantly poorer health status than the no-exposure group but were better than the dual-exposure group. These findings suggest that heat and noise exposure, particularly their co-exposure, may exert adverse effects on workers' health.

Association between heat/noise exposure and health outcomes

Figure 1 presents the results of multivariable logistic regression models. Model 1 is the unadjusted model, estimating crude associations between occupational exposures and health outcomes. Model 2 further adjusts for potential confounders, including age, gender, work duration, income level, protective measures, and company size. The findings were generally consistent across models. After adjustment, occupational heat exposure was

Table 2 Participant characteristics stratified by combined exposure categories

Variables	Overall	None	Heat only	Noise only	Heat and noise	*P*-value
	N = 10,275	*N* = 4330	*N* = 2107	*N* = 1903	*N* = 1935	
Age(years)	40.71 ± 10.08	40.14 ± 10.45	41.30 ± 9.65	40.78 ± 9.96	41.27 ± 9.75	< 0.001
Employment duration (years)	12.03 ± 9.89	10.86 ± 9.83	13.96 ± 10.46	11.72 ± 9.73	12.87 ± 9.16	< 0.001
Gender, n (%)						< 0.001
Male	8,652 (84.2%)	3,571 (82.5%)	1,799 (85.4%)	1,595 (83.8%)	1,687 (87.2%)	
Female	1,623 (15.8%)	759 (17.5%)	308 (14.6%)	308 (16.2%)	248 (12.8%)	
Income level, n (%)						< 0.001
Low	8,696 (84.6%)	3,689 (85.2%)	1,765 (83.8%)	1,619 (85.1%)	1,623 (83.9%)	
Medium	1,231 (12.0%)	481 (11.1%)	263 (12.5%)	198 (10.4%)	289 (14.9%)	
High	348 (3.4%)	160 (3.7%)	79 (3.7%)	86 (4.5%)	23 (1.2%)	
Company size, n (%)						< 0.001
Small	4,592 (45.4%)	1,757 (41.1%)	1,070 (51.2%)	1,041 (55.8%)	724 (38.2%)	
Large	5,533 (54.6%)	2,518 (58.9%)	1,019 (48.8%)	825 (44.2%)	1,171 (61.8%)	
Protect measure, n (%)						< 0.001
No	4,225 (41.1%)	1,664 (38.4%)	992 (47.1%)	654 (34.4%)	915 (47.3%)	
Yes	6,050 (58.9%)	2,666 (61.6%)	1,115 (52.9%)	1,249 (65.6%)	1,020 (52.7%)	
Hypertension, n (%)						< 0.001
No	7,347 (71.5%)	3,336 (77.0%)	1,427 (67.7%)	1,388 (72.9%)	1,196 (61.8%)	
Yes	2,928 (28.5%)	994 (23.0%)	680 (32.3%)	515 (27.1%)	739 (38.2%)	
Diabetes, n (%)						0.309
No	7,904 (92.7%)	3,081 (93.4%)	1,898 (92.6%)	1,260 (92.1%)	1,665 (92.2%)	
Yes	620 (7.3%)	219 (6.6%)	152 (7.4%)	108 (7.9%)	141 (7.8%)	
Renal dysfunction, n (%)						< 0.001
No	6,826 (81.8%)	2,919 (83.8%)	1,375 (78.7%)	1,196 (82.8%)	1,336 (80.2%)	
Yes	1,514 (18.2%)	563 (16.2%)	373 (21.3%)	249 (17.2%)	329 (19.8%)	
Liver dysfunction, n (%)						< 0.001
No	6,198 (72.1%)	2,551 (74.0%)	1,401 (69.9%)	1,039 (76.0%)	1,207 (67.7%)	
Yes	2,402 (27.9%)	894 (26.0%)	604 (30.1%)	328 (24.0%)	576 (32.3%)	
Pulmonary dysfunction, n (%)						0.006
No	4,993 (75.2%)	2,101 (77.2%)	977 (74.1%)	1,196 (75.0%)	719 (71.9%)	
Yes	1,643 (24.8%)	622 (22.8%)	342 (25.9%)	398 (25.0%)	281 (28.1%)	
Abnormal ECG, n (%)						< 0.001
No	7,297 (71.0%)	3,280 (75.8%)	1,498 (71.1%)	1,315 (69.1%)	1,204 (62.2%)	
Yes	2,978 (29.0%)	1,050 (24.2%)	609 (28.9%)	588 (30.9%)	731 (37.8%)	

Note: Table 2 presents the characteristics of workers stratified by combined heat and noise exposure, categorized into four groups: no exposure, heat only, noise only, and dual exposure

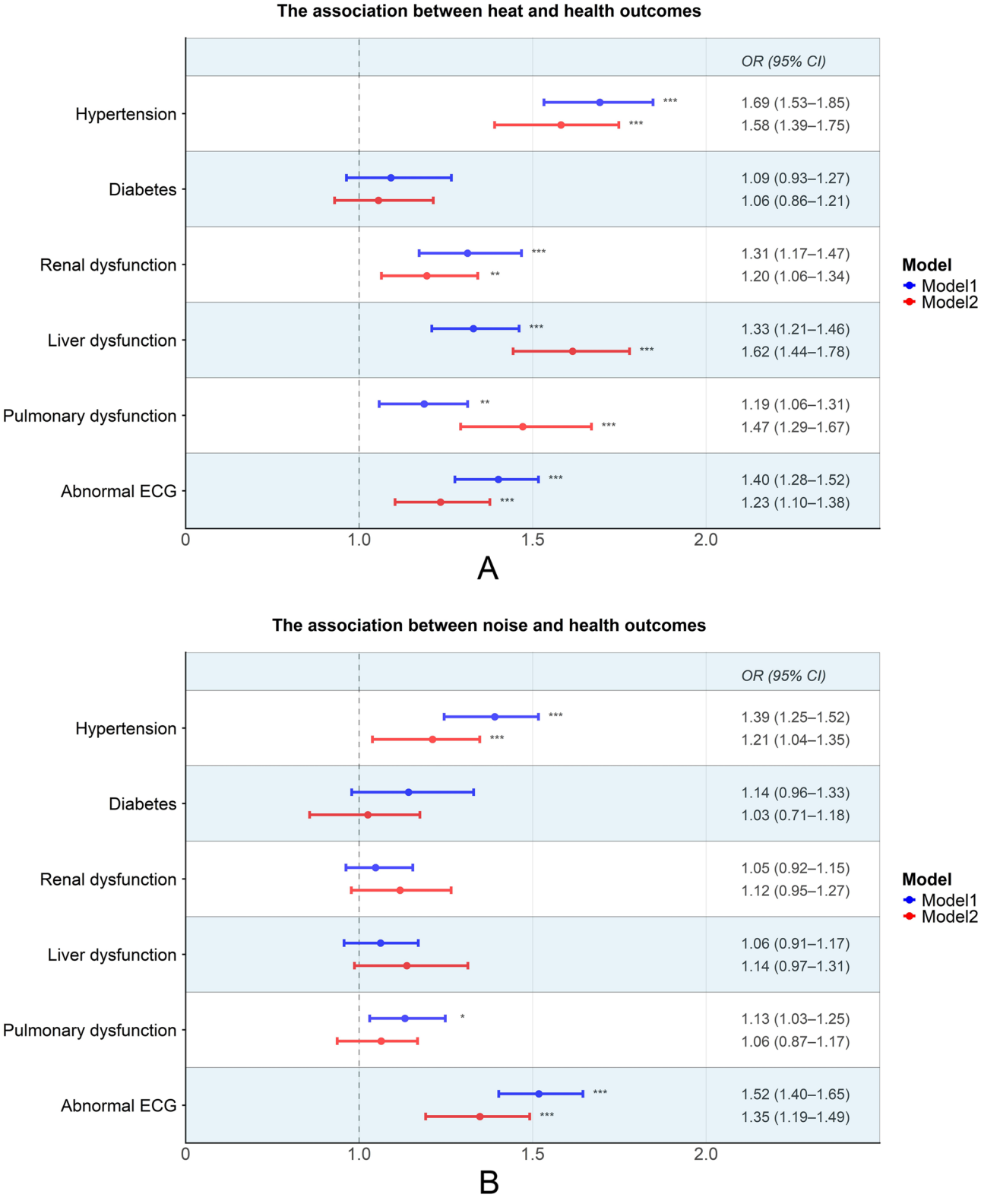


Fig. 1 Logistic regression analysis of exposures to heat and noise on different health outcomes. A: The association between heat and health outcomes. B: The association between noise and health outcomes. Model 1 was unadjusted; Model 2 was adjusted for age, gender, work duration, income level, protective measure, and company size. ***P< 0.001, **P< 0.01, *P < 0.05.

significantly associated with increased risks of hypertension (OR = 1.58, 95% CI: 1.39–1.75), renal dysfunction (OR = 1.20, 95% CI: 1.06–1.34), liver dysfunction (OR = 1.62, 95% CI: 1.44–1.78), pulmonary dysfunction (OR = 1.47, 95% CI: 1.29–1.67), and abnormal ECG (OR = 1.23, 95% CI: 1.10–1.38). No significant association was observed with diabetes. Similarly, noise exposure was significantly associated with hypertension (OR = 1.21, 95% CI: 1.04–1.35) and abnormal ECG (OR = 1.35, 95% CI: 1.19–1.49), but not with diabetes or renal dysfunction.

Interaction effects of heat and noise exposure

Figure 2 presents the results of RERI analysis examining the combined effects of heat and noise exposure on various health outcomes. The findings show that co-exposure to heat and noise significantly increased the risks of hypertension (OR=1.94, 95%CI: 1.61-2.36; RERI=0.26) and abnormal ECG (OR=1.81, 95%CI:1.48-2.12; RERI=0.16), with statistically significant interaction effects (*P* interaction < 0.001). RERI values of 0.26 and 0.16 suggest synergistic interactions, indicating that the excess relative risk due to interaction was 0.26 for hypertension and 0.16 for abnormal ECG. Furthermore, the analysis revealed that combined exposure was associated with increased risks of liver dysfunction (OR=1.58, 95%CI: 1.31-1.92; RERI=0.18) and pulmonary dysfunction (OR=1.38, 95%CI: 1.15-1.68; RERI=0.14), however, the interaction did not reach statistical significance. These results suggest that while concurrent heat and noise exposure may adversely affect multiple health outcomes, the synergistic effects are particularly evident in cardiovascular outcomes.

Prediction performance of random forest model for health outcomes

The predictive performance of the random forest model varied across different health outcomes, as shown in Table S1 and Figure S1. In terms of discriminative ability, all models achieved AUC values above 0.75, indicating good predictive performance. Pulmonary dysfunction (AUC=0.839, 95% CI: 0.819–0.860) and liver dysfunction (AUC=0.807, 95% CI: 0.786–0.827) demonstrated the

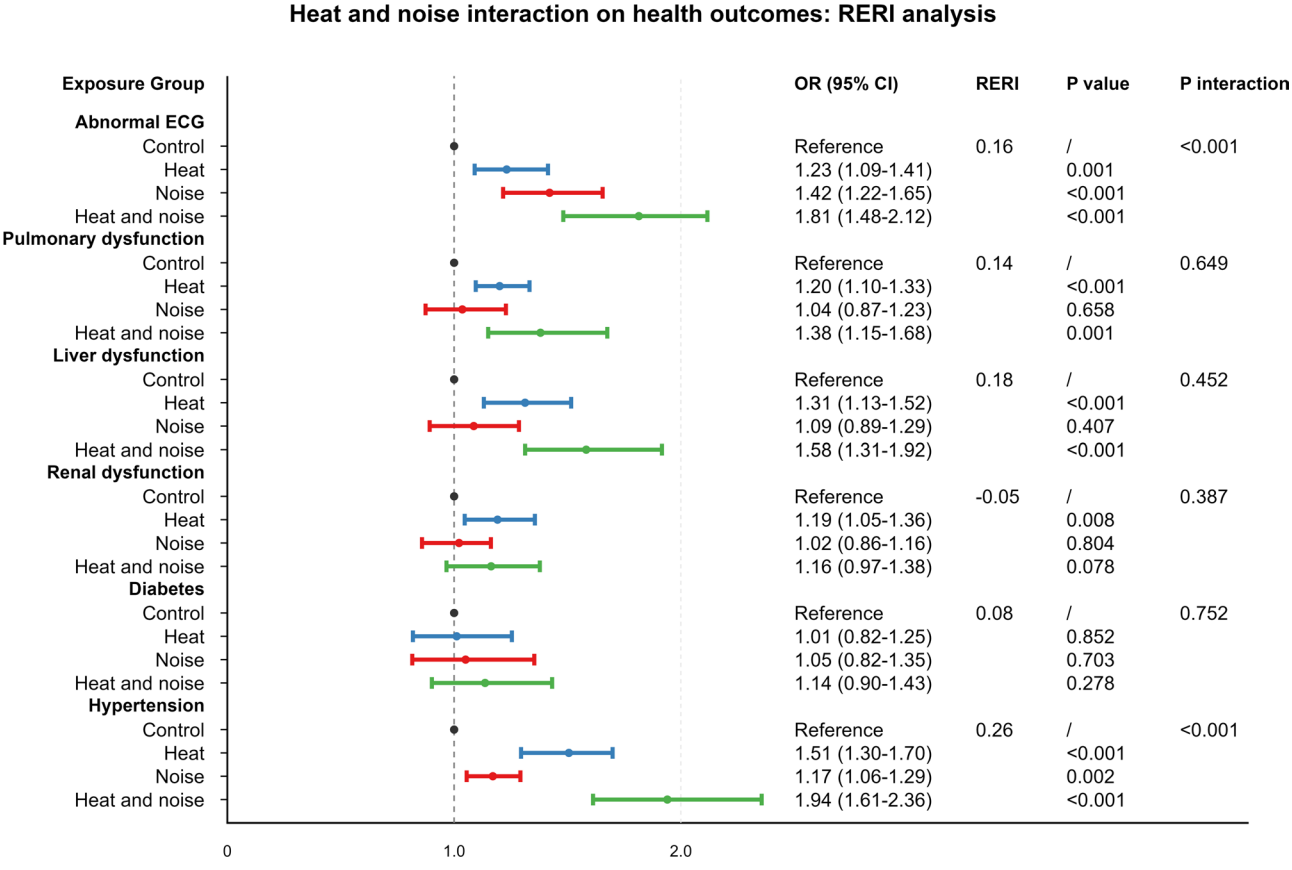


Fig. 2 Assessment of interaction effects between occupational heat and noise on health outcomes using RERI Analysis. The RERI was calculated as: RERI = OR11 - OR10 - OR01 + 1, where OR11 represents the odds ratio for joint exposure, OR10 and OR01 denote odds ratios for single exposures. A RERI >0 suggests synergistic interaction (combined effect exceeds sum of individual effects), RERI = 0 indicates no additive interaction, and RERI < 0 implies antagonistic interaction (combined effect weaker than individual effects sum). The model was adjusted for age, gender, employment duration, income level, protective measure, and company size

highest discriminative power, followed by renal dysfunction (AUC=0.784, 95% CI: 0.753–0.815) and abnormal ECG (AUC=0.771, 95% CI: 0.750–0.792). Classification metrics revealed that the model for renal dysfunction exhibited high accuracy (0.759), while liver dysfunction had the highest sensitivity (0.847), and renal dysfunction exhibited the highest specificity (0.771). Overall, the model demonstrated robust predictive performance across the evaluated health outcomes.

Analyzing the contribution of factors on occupational health outcomes using random forest and SHAP methods

Figure 3 and Table S2 show the mean absolute SHAP values derived from the random forest model, highlighting

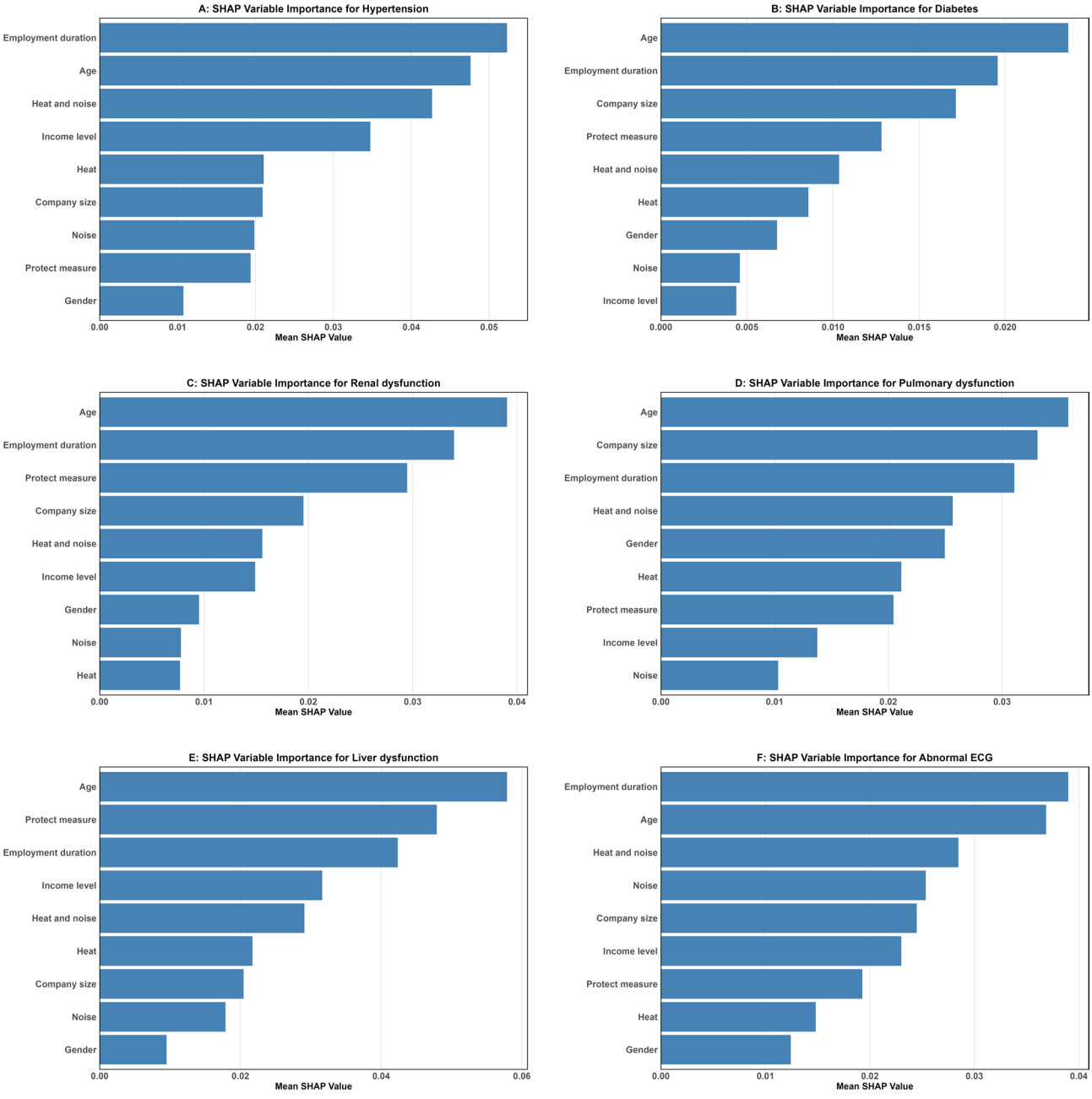


Fig. 3 Analysis of determinant factors for health outcomes using random forest and SHAP methods. A: SHAP Variable Importance for Hypertension. B: SHAP Variable Importance for Diabetes. C: SHAP Variable Importance for Renal dysfunction. D: SHAP Variable Importance for Pulmonary dysfunction. E: SHAP Variable Importance for Liver dysfunction. F: SHAP Variable Importance for Abnormal ECG. Random forest models (constructed using the R package randomForest) and SHAP (SHapley Additive exPlanations) methods were used to analyze determinants of health outcomes. SHAP values were computed using the R package fastshap, reflecting the marginal contribution of each feature to the model's predictive outcomes (based on the means of absolute SHAP values).

Table 3 Mediation of the association between occupational exposure and health outcomes by WBC

Exposure	Health outcome	Direct effect	Indirect effect	Total effect	Proportion of mediation
Heat	Hypertension	0.0579 (0.0360, 0.0802)***	0.0059 (0.0034, 0.0090)***	0.0638 (0.0418, 0.0869)***	9.2%
Heat	Diabetes	0.0033 (−0.0154, 0.0209)	0.0003 (−0.0004, 0.0012)	0.0036 (−0.0149, 0.0215)	8.3%
Heat	Renal dysfunction	0.0335 (0.0117, 0.0605)**	0.001 (−0.0009, 0.0017)	0.0345 (0.0131, 0.0608)**	2.9%
Heat	Pulmonary dysfunction	0.0523 (0.0169, 0.0875)***	0.0028 (−0.0003, 0.0065)	0.0550 (0.0220, 0.0888)***	5.1%
Heat	Liver dysfunction	0.0573 (0.0337, 0.0821)***	0.0033 (0.0015, 0.0053)***	0.0606 (0.0366, 0.0854)***	5.4%
Heat	Abnormal ECG	0.0526 (0.0199, 0.0819)***	0.0039 (0.0018, 0.0064)***	0.0565 (0.0239, 0.0861)***	7.0%
Noise	Hypertension	0.0225 (0.0032, 0.0291)***	0.0018 (0.0009, 0.0025)***	0.0243 (0.0036, 0.0316)***	7.4%
Noise	Diabetes	0.0041 (−0.0105, 0.0198)	0.0007 (−0.0014, 0.0027)	0.0048 (−0.0087, 0.0196)	14.6%
Noise	Renal dysfunction	0.0097 (−0.0173, 0.0344)	0.0004 (−0.0006, 0.0013)	0.0101 (−0.0266, 0.0364)	3.9%
Noise	Pulmonary dysfunction	0.0169 (−0.0205, 0.0512)	0.0002 (−0.0016, 0.0007)	0.0171 (−0.0213, 0.0505)	1.2%
Noise	Liver dysfunction	0.0217 (−0.0090, 0.0547)	0.0033 (0.0013, 0.0057)***	0.0250 (−0.0059, 0.0585)	13.2%
Noise	Abnormal ECG	0.0726 (0.0517, 0.0952)***	0.0049 (0.0025, 0.0078)***	0.0775 (0.0561, 0.1006)***	6.4%
Heat and noise	Hypertension	0.0791 (0.0497, 0.1087)***	0.0143 (0.0105, 0.0185)***	0.0934 (0.0660, 0.1220)***	15.3%
Heat and noise	Diabetes	0.0085 (−0.0052, 0.0231)	0.0003 (−0.0004, 0.0009)	0.0088 (−0.0052, 0.0234)	3.4%
Heat and noise	Renal dysfunction	0.0123 (−0.0108, 0.0387)	0.0010 (−0.0034, 0.0054)	0.0133 (−0.0092, 0.0386)	7.6%
Heat and noise	Pulmonary dysfunction	0.0603 (0.0178, 0.1052)*	0.0054 (−0.0047, 0.0163)	0.0657 (0.0248, 0.1083)**	8.2%
Heat and noise	Liver dysfunction	0.0631 (0.0367, 0.0919)***	0.0082 (0.0041, 0.0116)***	0.0713 (0.0449, 0.0998)***	11.5%
Heat and noise	Abnormal ECG	0.0744 (0.0449, 0.0951)***	0.0071 (0.0027, 0.0109)***	0.0815 (0.0563, 0.1065)***	8.7%

Note: Mediation analysis was performed using R's mediation package to assess WBC's role in occupational exposure and health outcomes. All models were adjusted for age, gender, employment duration, income level, company size and protect measure. Effects were estimated via 10,000 Monte Carlo simulations, with direct effects (ADE), indirect effects (ACME), total effects, and mediation proportions reported

*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$

the relative contribution of key predictors across various health outcomes. Age and employment duration consistently emerged as the two most influential predictors, ranking highest for hypertension (SHAP: 0.048 and 0.052, respectively), liver dysfunction (0.058 and 0.042) and abnormal ECG (0.037 and 0.039). A noteworthy finding was the synergistic effect of combined heat and noise exposure, which demonstrated a greater contribution than single exposure alone across all health outcomes, underscoring the importance of evaluating interactive effects in occupational health assessments.

Mediation analysis of WBC's role in occupational exposures and health outcomes

Table S3 systematically compares the distribution characteristics of WBC across different occupational exposure combinations. The data reveal a significant exposure gradient in WBC levels: the non-exposure group showed the lowest values ($6.38\pm2.80\times10^9$/L), followed by single-exposure groups (heat-only: $6.76\pm2.62\times10^9$/L; noise-only: $6.63\pm1.80\times10^9$/L), while the heat and noise co-exposure group exhibited the highest levels ($8.32\pm4.34\times10^9$/L, $P<0.001$). Notably, WBC levels were further elevated in patients with various chronic diseases compared to healthy individuals, with the most pronounced increases observed in the co-exposure group. For instance, patients with hypertension ($10.17\pm4.90\times10^9$/L), liver dysfunction ($11.33\pm4.79\times10^9$/L) and abnormal ECG findings ($10.23\pm4.65\times10^9$/L) in the co-exposure group exceeded normal WBC ranges (4-10.0×10^9/L) and showed significantly higher values than single-exposure or non-exposure groups (all $P<0.001$).

Mediation analysis in Table 3 further elucidates the Mediating role of WBC between occupational exposures and health outcomes. For single heat exposure, WBC significantly mediated associations with hypertension (mediation proportion 9.2%), liver dysfunction (5.4%), and abnormal ECG findings (7.0%) (all $P<0.001$). Under single noise exposure, WBC also demonstrated significant mediation effects for hypertension (7.4%) and abnormal ECG findings (6.4%) ($P<0.001$). Remarkably, during heat and noise co-exposure, the mediating role of WBC was enhanced, with mediation proportions increasing to 15.3% for hypertension and 8.7% for abnormal ECG findings (all $P<0.001$). Meanwhile, WBC also has a mediating effect on liver dysfunction in the co-exposure group (11.5%, $P<0.001$). These findings demonstrate that inflammation-related changes, as represented by WBC count, play quantitatively distinct roles in connecting occupational exposures with adverse health outcomes under different exposure conditions.

Sensitivity analysis addressing healthy worker effect bias

To assess the potential bias of the Healthy Worker Effect (HWE), we additionally included data from 311 workers who left employment with complete health examination records and occupational exposure profiles during 2020-2022. After inclusion, the prevalence rates of all health outcomes increased: hypertension rose from 28.5% to 29.6%, diabetes from 7.3% to 7.6%, renal dysfunction from 18.2% to 19.1%, pulmonary dysfunction from 24.8% to 25.4%, liver dysfunction from 27.9% to 29.2%, and abnormal ECG from 29.0% to 30.3%. The sensitivity analysis results in Table S4 revealed increases in OR across most exposure categories compared to the original analysis in Table 1. For instance, the OR of for hypertension for single heat exposure increased from 1.51 (1.30-1.70) to 1.64 (1.38-2.02), for single noise exposure increased from 1.17 (1.06-1.29) to 1.28 (1.13-1.49), while for heat and noise co-exposure rose from 1.94 (1.61-2.36) to 2.11 (1.76-2.83), similar patterns were observed for abnormal ECG, renal dysfunction, pulmonary dysfunction and liver dysfunction. These findings confirm the existence of HWE. Nevertheless, the statistical significance of the association between combined heat and noise exposure and health outcomes remained unchanged or even increased, indicating that the overarching conclusion on occupational health risks remains robust.

Discussion

This study, based on a multi-industry cross-sectional survey covering all prefecture-level cities in Fujian Province, China, investigated the effects of combined workplace heat and noise exposure on multi-system health outcomes among Chinese occupational populations. The results not only reaffirmed the health risks associated with single exposures but also revealed synergistic effects of combined heat and noise exposure and their underlying inflammatory mediation mechanisms. These findings provide important evidence for workplace hazard risk assessment and the development of protective measures, and hold significant implications for mitigating heat-related health burdens in the context of climate warming [32].

This study identified significant associations between occupational heat exposure and multiple health risks, including hypertension, abnormal ECG findings, renal dysfunction, liver dysfunction, and pulmonary dysfunction. These findings are consistent with previous international studies. For instance, a cohort study of Costa Rican rice farmers demonstrated that chronic heat exposure led to decreased estimated glomerular filtration rate (eGFR), suggesting that heat stress may cause irreversible kidney damage [33]. Similarly, an observational study of U.S. firefighters revealed significantly increased incidence of coronary syndrome and abnormal ECG findings during fire suppression activities, further corroborating the cardiovascular risks of occupational heat exposure [34]. From a pathophysiological perspective, the mechanisms underlying heat-induced multi-system damage may involve several pathways. Heat stress can activate the sympathetic nervous system, leading to increased cardiac output and peripheral vascular resistance, which subsequently induces sustained blood pressure elevation and myocardial remodeling. Additionally, heat-related dehydration may trigger the renin-angiotensin-aldosterone system (RAAS), simultaneously increasing cardiovascular load and causing direct structural damage to renal tubules [35–37]. These complex pathophysiological mechanisms collectively form the scientific basis for heat exposure-induced multi-system dysfunction.

Noise exposure is a well-recognized occupational hazard affecting both auditory and non-auditory systems. Chronic noise exposure can lead to a spectrum of health impairments ranging from sleep disturbances and cognitive dysfunction to metabolic disorders and increased diabetes risk [38, 39]. Our findings further substantiate this body of evidence, demonstrating that occupational noise exposure elevates risks of hypertension and abnormal ECG findings. These results align with global epidemiological data, including comprehensive reviews reporting multifactorial and complex relationships between built environments and increased cardiovascular disease risk with poor prognosis [40, 41]. Furthermore, a systematic review and meta-analysis examining cardiovascular health effects of noise exposure documented impacts on blood pressure, hypertension, heart rate, arrhythmias, vascular resistance, and cardiac output at higher exposure levels [42]. The pathophysiological mechanisms underlying these noise-induced health effects likely involve multiple biological pathways, including oxidative stress activation, induction of systemic inflammatory responses, and prothrombotic state formation [43, 44].

This study advances current knowledge by revealing significant synergistic effects of combined heat and noise exposure through both traditional and machine learning models. Specifically, logistic regression analyses and RERI analysis demonstrated more pronounced hypertension and ECG abnormality risks under dual exposure compared to single exposures. This synergistic effect is supported by experimental evidence: Dehghan et al.[45] confirmed in climate chamber studies that heat exposure significantly exacerbates noise-induced blood pressure elevation. Other studies have documented similar worsening trends in blood pressure variability under combined exposure conditions [46]. An experimental study involving 216 elderly male participants in Iran further showed that while heart rate increased with rising noise levels, this effect became significantly more pronounced

at higher ambient temperatures, even with relatively low noise levels [47]. Similarly, a simulation of neurophysiological responses to combined noise and heat in indoor environments revealed that elevated temperatures amplified neurophysiological responses more substantially when accompanied by high noise levels, with more marked combined effects on heart rate and respiratory frequency than either factor alone [48].

Subsequently, our random forest machine learning model identified the interaction between heat and noise exposure as a more significant predictor of adverse health outcomes than either exposure alone. These findings reinforce the importance of addressing combined occupational hazards rather than managing them in isolation and have practical implications for workplace interventions. First, priority should be given to identifying and mitigating risks in work environments where simultaneous heat and noise exposures are prevalent. This includes implementing joint monitoring systems for temperature, humidity, and noise levels to enable real-time hazard assessment. Engineering controls—such as installing soundproof cooling equipment, optimizing ventilation, and using noise-absorbing materials alongside localized cooling—can help reduce dual exposure levels. Operational strategies should also consider rescheduling high-noise tasks to cooler periods of the day to minimize combined stress. Additionally, occupational safety training programs should incorporate the risks associated with co-exposure to heat and noise, aiming to raise worker awareness and improve the effectiveness of protective behaviors. These integrated measures offer a more robust framework for safeguarding worker health in complex exposure scenarios.

Through mediation analysis, this study found that WBC plays a mediating role in the relationship between occupational exposure and multiple health outcomes. Although the effect sizes varied, these results highlight the critical role of systemic inflammation as a common underlying mechanism for cardiovascular, hepatic, and respiratory dysfunction [49–51]. In the study, we stratified the population's WBC levels based on coexposure status and disease conditions, revealing that in cases where coexposure and disease coexisted, WBC levels not only exceeded the normal range but were also significantly higher than those in the non-exposed group or single-exposure group. Combined with the mediation analysis results, WBC exhibited a higher mediating proportion in the association between coexposure and disease, suggesting that the synergistic effect of combined exposure may amplify inflammatory responses and exacerbate disease progression. Mechanistically, long-term heat exposure may lead to persistent immune activation, triggering elevated WBC counts, which in turn contribute to endothelial dysfunction and myocardial fibrosis, ultimately resulting in disease onset [52]. Meanwhile, noise exposure can stimulate the proliferation of inflammatory cells such as WBC through mechanisms involving the hypothalamic-pituitary-adrenal axis [53]. Therefore, when both factors act together, they may produce an additive inflammatory activation effect, which could explain the higher mediating proportion of WBC under coexposure conditions. In this study, although the WBC levels in the single-exposure group did not exceed the normal range, they were still significantly higher than those in the non-exposed group. Such an elevation within the normal range remains epidemiologically meaningful, suggesting that some workers may already exhibit a subclinical inflammatory state, characterized by low-grade systemic inflammation despite the absence of overt clinical symptoms. A meta-analysis has indicated that even within the normal range, WBC levels maintain a statistically significant association with disease incidence [54]. Similarly, multiple previous studies have demonstrated that even modest increases in WBC within the normal range can still mediate the relationship between exposure and disease [55–63]. Therefore, these subtle changes in WBC may reflect early subclinical damage, serving as an early warning sign for intervention in occupational health monitoring.

Furthermore, although the mediating proportion of WBC in some single-exposure outcomes was numerically small, its public health significance should not be overlooked. These "small effects" may be part of a broader, multifactorial pathogenic network. Occupational exposures often impact health through multiple, interacting pathways, and even modest mediation effects can contribute substantially when considered cumulatively [64–66]. Beyond systemic inflammation reflected by WBC counts, other biological pathways may also mediate the association between heat or noise exposure and disease. Animal studies have shown that heat stress can induce respiratory diseases through various mediators, including serum immune markers (Total-IgE, OVA-sIgE, and IL-4), pro-inflammatory cytokines (IL-6 and TNF-α), and oxidative stress markers (8-OHdG, ROS, and MDA) [67]. Meanwhile, a Mendelian randomization study examining noise as the exposure, 91 inflammatory factors as mediators, and 26 diseases as outcomes revealed that noise exposure could contribute to disease development through multiple inflammatory mediators, including cytokines, chemokines, and prostaglandins [68]. Under conditions of co-exposure, these mechanisms interact, collectively forming a complex pathogenic network in occupational settings. These findings not only deepen our understanding of the mechanisms underlying occupational exposure-related diseases but, more importantly, provide a solid theoretical foundation for developing

multi-target intervention strategies centered on modulating inflammatory responses.

Finally, although this study was conducted in Fujian Province, China, the findings offer valuable reference points for other regions and countries. As a major coastal industrial province in southeastern China, Fujian shares comparable industrial structures (e.g., manufacturing and mining) with many industrialized regions globally, supporting the broader applicability of the results [69, 70]. The occupational exposure limits used in this study are based on national standards in China, while the diagnostic criteria for major health outcomes adhere to internationally recognized guidelines—both of which strengthen the external validity of our findings. Nonetheless, we acknowledge that variations in climate conditions, worker acclimatization, and occupational health policies may influence the degree to which our conclusions apply elsewhere. For example, workers in tropical climates may develop physiological adaptations to heat, whereas those in temperate or colder environments may exhibit increased vulnerability to thermal stress [71–73]. Furthermore, differences in occupational safety standards, enforcement, and access to protective measures must be considered when applying these findings in other settings. Such contextual factors should be carefully evaluated to ensure appropriate interpretation and implementation of our results across diverse occupational environments.

Limitations

This study has several limitations. First, its cross-sectional design allows for the identification of associations between exposures and health outcomes but precludes any inference of causality. Second, although data from workers who left employment during the survey period were included in sensitivity analyses, the lack of detailed information regarding their specific reasons for leaving may introduce residual HWE bias. Third, while major confounders were adjusted, unmeasured lifestyle factors like smoking status, alcohol consumption, and dietary patterns may still affect results. Furthermore, the lack of specific inflammatory biomarkers (e.g., hs-CRP, IL-6, TNF-α, and fibrinogen) limits mechanistic insights.

To address these limitations, future research should consider adopting longitudinal cohort designs to track changes in exposure and health outcomes over time. Complementary controlled experiments (e.g., intervention or quasi-experimental studies) could provide further causal validation. Moreover, improvements in occupational health recordkeeping—particularly regarding job exit reasons—would help refine assessments of the HWE. Finally, systematic collection of lifestyle data and relevant biomarkers would enhance adjustment for confounding factors and strengthen the biological plausibility of observed associations.

Conclusion

This study provides evidence that concurrent exposure to occupational heat and noise exerts synergistic effects, significantly amplifying multi-system health risks in the Chinese working population. Systemic inflammation, as indicated by WBC count, appears to be a key mediating mechanism. These findings challenge the adequacy of traditional single-exposure assessment models and underscore the urgent need for integrated evaluation of combined occupational hazards. Our results offer a scientific foundation for developing more precise, inflammation-focused occupational health prevention and intervention strategies.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12889-025-24592-1.

Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

Supplementary Material 4.

Acknowledgements

The authors would like to express their gratitude to the Fujian Center for Prevention and Control of Occupational Diseases and Chemical Poisoning for the provision of the surveillance data.

Authors' contributions

M.L., Y.Z. and J.X. designed the study. Y.Z., C.W., Y.J., and J.L. in collected the data. M.L., J.X. and J.L. analysed the data. Y.Z.,C.W. and Y.J. contributed to the interpretation of results. M.L., J.X. and J.L. wrote and reviewed the manuscript. All authors discussed the results and commented on the manuscript. All authors read and approved the final manuscript.

Funding

This work was supported by the Minjiang Scholar Start-up Research Fund of Fujian Province (Grant No. 2019-9202001001).

Data availability

The datasets used in this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

This study was approved by the Ethics Committee of Fujian Medical University (Approval No. 2022 – 111).

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 13 June 2025 / Accepted: 25 August 2025
Published online: 21 October 2025

References

1. He J, Smith LM, Shkembi A, Neitzel RL. Evaluating the impact of occupational noise exposure on workplace fatal and nonfatal injuries in the U.S. (2006–2020). Int J Hyg Environ Health. 2025;263:114468.
2. Zhang Y, Chen Y, Su Q, Huang X, Li Q, Yang Y, et al. The use of machine and deep learning to model the relationship between discomfort temperature and labor productivity loss among petrochemical workers. BMC Public Health. 2024;24(1):3269.
3. Han S, Dong L, Weng Y, Xiang J. Heat exposure and productivity loss among construction workers: a meta-analysis. BMC Public Health. 2024;24(1):3252.
4. Gedik Toker Ö, Tas Elibol N, Kuru E, Görmezoğlu Z, Görener A, Toker K. Industrial noise: impacts on workers' health and performance below permissible limits. BMC Public Health. 2025;25(1):1615.
5. Xu NS. Z: Heat and health of occupational workers: a short summary of literature. J Occup Health. 2024;66(1):uiae018.
6. Flouris AD, Dinas PC, Ioannou LG, Nybo L, Havenith G, Kenny GP, Kjellstrom T. Workers' health and productivity under occupational heat strain: a systematic review and meta-analysis. Lancet Planet Health. 2018;2(12):e521–31.
7. Li W, Zhu J, Hao X, Chen Z, Wang D. Noise exposure in the workplace, genetic susceptibility, and incidence of atrial fibrillation: a prospective cohort study. J Am Heart Assoc. 2024;13(21):e036543.
8. Teixeira LR, Pega F, Dzhambov AM, Bortkiewicz A, da Silva DTC, de Andrade CAF, Gadzicka E, Hadkhale K, Iavicoli S, Martínez-Silveira MS, et al. The effect of occupational exposure to noise on ischaemic heart disease, stroke and hypertension: A systematic review and meta-analysis from the WHO/ILO joint estimates of the Work-Related burden of disease and injury. Environ Int. 2021;154:106387.
9. López-Bueno JA, Padrón-Monedero A, Díaz J, Navas-Martín MA, Linares C. Short-term impact of air pollution, noise and temperature on emergency hospital admissions in Madrid (Spain) due to liver and gallbladder diseases. Environ Res. 2024;249:118439.
10. Blaustein JR, Quisel MJ, Hamburg NM, Wittkopp S. Environmental impacts on cardiovascular health and biology: an overview. Circ Res. 2024;134(9):1048–60.
11. Guo J, Garshick E, Si F, Tang Z, Lian X, Wang Y, et al. Environmental toxicant exposure and depressive symptoms. JAMA Netw Open. 2024;7(7):e2420259.
12. Wang Y, Meng Z, Wei S, Li X, Su Z, Jiang Y, et al. Urinary volatile organic compound metabolites and COPD among US adults: mixture, interaction and mediation analysis. Environ Health. 2024;23(1):45.
13. Fu Z, Zhou S, Weng Z, Liang G, Gu A. Association between nickel and multiple metabolic outcomes: the mediating roles of lipid metabolism and inflammation. Ecotoxicol Environ Saf. 2025;298:118274.
14. Jafari MJ, Pirposhteh EA, Dehghan SF, Khodakarim S, Jafari M. Relationship between heat stress exposure and some immunological parameters among foundry workers. Int J Biometeorol. 2020;64(5):853–61.
15. Gao Y, Liu Y, He J, Zhang Y, Wang T, Wu L, Sun N, Fang T, Mao H, Tang NJ, et al. Effects of heat waves and cold spells on blood parameters: a cohort study of blood donors in tianjin, China. Environ Health Prev Med. 2024;29:25.
16. Barros B, Oliveira M, Morais S. Firefighters' occupational exposure: contribution from biomarkers of effect to assess health risks. Environ Int. 2021;156:106704.
17. Mohammadi H, Alimohammadi I, Roshani S, Pakzad R, Abdollahi MB, Dehghan SF. The effect of occupational noise exposure on blood and biochemical parameters: a case study of an insulator manufacturer in Iran. Electron Physician. 2016;8(1):1740–6.
18. China, MoHotPsRo. GBZT189.7-2007-Measurement of physical factors in the workplace part 7: high temperature. In. Beijing: China Standard; 2007.
19. National Health Commission of the People's Republic of China. Classification of occupational hazards at workplaces – Part 4: Occupational exposure to noise. Beijing: National Health Commission of the People's Republic of China; 2012.
20. Currie KD, Schultz MG, Millar PJ, Pescatello LS. The role of exercise blood pressure in hypertension: measurement, mechanisms, and management. Med Sci Sports Exerc. 2025;57(2):425–33.
21. Pickering TG, Hall JE, Appel LJ, Falkner BE, Graves J, Hill MN, et al. Recommendations for blood pressure measurement in humans and experimental animals: Part 1: blood pressure measurement in humans: a statement for professionals from the subcommittee of professional and public education of the American heart association Council on high blood pressure research. Circulation. 2005;111(5):697–716.
22. Ma H, Liu F, Li J, Chen J, Cao J, Chen S, et al. Sex differences in associations between socioeconomic status and incident hypertension among Chinese adults. Hypertension. 2023;80(4):783–91.
23. McCarthy CP, Bruno RM, Rahimi K, Touyz RM, McEvoy JW. What is new and different in the 2024 European society of cardiology guidelines for the management of elevated blood pressure and hypertension?? Hypertension. 2025;82(3):432–44.
24. ElSayed NA, Aleppo G, Aroda VR, Bannuru RR, Brown FM, Bruemmer D, Collins BS, Cusi K, Das SR, Gibbons CH, et al. Summary of revisions: standards of care in Diabetes-2023. Diabetes Care. 2023;46(Suppl 1):S5–9.
25. Levey AS, Eckardt KU, Tsukamoto Y, Levin A, Coresh J, Rossert J, et al. Definition and classification of chronic kidney disease: a position statement from kidney disease: improving global outcomes (KDIGO). Kidney Int. 2005;67(6):2089–100.
26. Levey AS, Atkins R, Coresh J, Cohen EP, Collins AJ, Eckardt KU, et al. Chronic kidney disease as a global public health problem: approaches and initiatives - a position statement from kidney disease improving global outcomes. Kidney Int. 2007;72(3):247–59.
27. Graham BL, Steenbruggen I, Miller MR, Barjaktarevic IZ, Cooper BG, Hall GL, Hallstrand TS, Kaminsky DA, McCarthy K, McCormack MC, et al. Standardization of spirometry 2019 update. An official American thoracic society and European respiratory society technical statement. Am J Respir Crit Care Med. 2019;200(8):e70–88.
28. Quanjer PH, Stanojevic S, Cole TJ, Baur X, Hall GL, Culver BH, et al. Multi-ethnic reference values for spirometry for the 3-95-yr age range: the global lung function 2012 equations. Eur Respir J. 2012;40(6):1324–43.
29. Xu J, Xu ZX, Yang QF, Zhuang J, Zhu X, Yao J. Association between the sarcopenia index and abnormal liver function in the adult population in the United States: a cross-sectional study. Front Med. 2023;10:1266253.
30. Kim WD, Lee Y, Kim BS, Kim HJ, Shin JH, Park JK, et al. Electrocardiography score based on the Minnesota code classification system predicts cardiovascular mortality in an asymptomatic low-risk population. Ann Med. 2023;55(2):2288306.
31. Kligfield P, Gettes LS, Bailey JJ, Childers R, Deal BJ, Hancock EW, et al. Recommendations for the standardization and interpretation of the electrocardiogram: part I: the electrocardiogram and its technology: a scientific statement from the American Heart Association Electrocardiography and Arrhythmias Committee, Council on Clinical Cardiology; the American College of Cardiology Foundation; and the Heart Rhythm Society: endorsed by the International Society for Computerized Electrocardiology. Circulation. 2007;115(10):1306–24.
32. Ensuring safety. and health at work in a changing climate [https://www.ilo.org/publications/ensuring-safety-and-health-work-changing-climate]
33. Crowe J, Rojas-Valverde D, Rojas-Garbanzo M, Gutiérrez-Vargas R, Ugalde-Ramírez JA, Ledezma-Rojas JP, Cabrera-Alpizar W, Salazar-Salazar M, Mauricio-La Torre R, Valera-Amador L et al. Kidney function in rice workers exposed to heat and dehydration in Costa Rica. Int J Environ Res Public Health. 2022;19(9):4962.
34. Al-Zaiti S, Rittenberger JC, Reis SE, Hostler D. Electrocardiographic responses during fire suppression and recovery among experienced firefighters. J Occup Environ Med. 2015;57(9):938–42.
35. Inoue Y, Ichinose-Kuwahara T, Nakamura S, Ueda H, Yasumatsu H, Kondo N, et al. Cutaneous vasodilation response to a linear increase in air temperature from 28 degrees C to 40 degrees C in prepubertal boys and young men. J Physiol Anthropol. 2009;28(3):137–44.
36. Cheng X, Su H. Effects of climatic temperature stress on cardiovascular diseases. Eur J Intern Med. 2010;21(3):164–7.
37. Shan R, Liu N, Yan Y, Liu B. Apoptosis, autophagy and atherosclerosis: relationships and the role of Hsp27. Pharmacol Res. 2021;166:105169.
38. Hammer MS, Swinburn TK, Neitzel RL. Environmental noise pollution in the United States: developing an effective public health response. Environ Health Perspect. 2014;122(2):115–9.
39. Passchier-Vermeer W, Passchier WF. Noise exposure and public health. Environ Health Perspect. 2000;108(Suppl 1):123–31.
40. Noorali AA, Hussain Merchant AA, Afzal N, Sen R, Junaid V, Khoja A, et al. Built environment and cardiovascular diseases - insights from a global review. Curr Atheroscler Rep. 2025;27(1):36.
41. Liu M, Meijer P, Lam TM, Timmermans EJ, Grobbee DE, Beulens JWJ, et al. The built environment and cardiovascular disease: an umbrella review and meta-meta-analysis. Eur J Prev Cardiol. 2023;30(16):1801–27.
42. Sivakumaran K, Ritonja JA, Waseem H, AlShenaibar L, Morgan E, Ahmadi SA, Denning A, Michaud DS, Morgan RL. Impact of noise exposure on risk of

developing Stress-Related health effects related to the cardiovascular system: A systematic review and Meta-Analysis. Noise Health. 2022;24(114):107–29.
43. Münzel T, Sørensen M, Gori T, Schmidt FP, Rao X, Brook FR, et al. Environmental stressors and cardio-metabolic disease: part II-mechanistic insights. Eur Heart J. 2017;38(8):557–64.
44. Herzog J, Schmidt FP, Hahad O, Mahmoudpour SH, Mangold AK, Garcia Andreo P, et al. Acute exposure to nocturnal train noise induces endothelial dysfunction and pro-thromboinflammatory changes of the plasma proteome in healthy subjects. Basic Res Cardiol. 2019;114(6):46.
45. Dehghan H, Bastami MT, Mahaki B. Evaluating combined effect of noise and heat on blood pressure changes among males in climatic chamber. J Educ Health Promot. 2017;6:39.
46. Jafari MJ, Khosrowabadi R, Khodakarim S, Khodagholi F, Mohammadian F. The effects of combined exposure to noise and heat on human salivary cortisol and blood pressure. Int J Occup Saf Ergon. 2021;27(3):831–9.
47. Tohidi M, Fatahi K, Beigi M. Investigating the effect of ambient temperature, sound type, and noise level on the heart rate of the elderly. Exp Gerontol. 2025;208:112829.
48. Abbasi AM, Motamedzade M, Aliabadi M, Golmohammadi R, Tapak L. Combined effects of noise and air temperature on human neurophysiological responses in a simulated indoor environment. Appl Ergon. 2020;88:103189.
49. Farrell SW, Leonard D, Shuval K, DeFina LF, Barlow CE, Pavlovic A, et al. Cardiorespiratory fitness, white blood cell count, and mortality in men and women. J Sport Health Sci. 2022;11(5):605–12.
50. Verma N, Roy A, Valsan A, Garg P, Ralmilay S, Girish V, et al. Liver dysfunction and systemic inflammation drive organ failures in acute decompensation of cirrhosis: a multicentric study. Am J Gastroenterol. 2025;120(1):182–93.
51. Yang HF, Kao TW, Wang CC, Peng TC, Chang YW, Chen WL. Serum white blood cell count and pulmonary function test are negatively associated. Acta Clin Belg. 2015;70(6):419–24.
52. Zhang Y, Hong S, Zhang Q, Su Z, Hu G, Wang T, Yu S, Yuan F, Zhu X, Jia G. Relationship between systemic inflammation and lung injury induced by chromate exposure: A cross-sectional study in workers. J Hazard Mater. 2023;452:131294.
53. She X, Gao X, Wang K, Yang H, Ma K, Cui B, et al. Effects of noise and low-concentration carbon monoxide exposure on rat immunity. J Occup Health. 2021;63(1):e12235.
54. Raya-Cano E, Vaquero-Abellán M, Molina-Luque R, Molina-Recio G, Guzmán-García JM, Jiménez-Mérida R, et al. Association between metabolic syndrome and leukocytes: systematic review and meta-analysis. J Clin Med. 2023. https://doi.org/10.3390/jcm12227044.
55. Liu Y, Lai X, Guo W, Ma L, Li W, Fang Q, et al. Total white blood cell count mediated the association between increased arterial stiffness and risk of type 2 diabetes mellitus in Chinese adults. Arterioscler Thromb Vasc Biol. 2020;40(4):1009–15.
56. Xu B, Wu Q, La R, Lu L, Abdu FA, Yin G, et al. Is systemic inflammation a missing link between cardiometabolic index with mortality? Evidence from a large population-based study. Cardiovasc Diabetol. 2024;23(1):212.
57. Torrungruang K, Vathesatogkit P, Mahanonda R, Thienpramuk L. Periodontitis and hypertension are linked through systemic inflammation: a 5-year longitudinal study. J Clin Periodontol. 2024;51(5):536–46.
58. Liu F, Li Y, Li W, Feng R, Zhao H, Chen J, et al. The role of peripheral white blood cell counts in the association between central adiposity and glycemic status. Nutr Diabetes. 2024;14(1):30.
59. Meng X, Han L, Fu J, Hu C, Lu Y. Associations between metabolic syndrome and depression, and the mediating role of inflammation: based on the NHANES database. J Affect Disord. 2025;375:214–21.
60. La R, Yin Y, Ding W, He Z, Lu L, Xu B, et al. Is inflammation a missing link between relative handgrip strength with hyperlipidemia? Evidence from a large population-based study. Lipids Health Dis. 2024;23(1):159.
61. Li C, Xiang S. Adverse childhood experiences, inflammation, and depressive symptoms in late life: a population-based study. J Gerontol B Psychol Sci Soc Sci. 2023;78(2):220–9.
62. Wan Z, Song L, Hu L, Lei X, Huang Y, Lv Y, et al. The role of systemic inflammation in the association between serum 25-hydroxyvitamin D and type 2 diabetes mellitus. Clin Nutr. 2021;40(6):3661–7.
63. Sun X, Zhu J, Qian Z, Chen X, Zhang J, Ji C, et al. A population-based study of the mediating role of WBC, NEUT and PLT in the relationship between Triglyceride-Glucose index and urinary albumin excretion. J Inflamm Res. 2024;17:10613–26.
64. Hanahan D, Weinberg RA. Hallmarks of cancer: the next generation. Cell. 2011;144(5):646–74.
65. Böhnke JR. Explanation in causal inference: methods for mediation and interaction. Q J Exp Psychol (Hove). 2016;69(6):1243–4.
66. Chen Y, Wu Y, Lv J, Zhou S, Lin S, Huang S, et al. Overall and individual associations between per- and polyfluoroalkyl substances and liver function indices and the metabolic mechanism. Environ Int. 2024;183:108405.
67. Lu C, Liu Q, Deng M, Liao H, Yang X, Ma P. Interaction of high temperature and NO(2) exposure on asthma risk: in vivo experimental evidence of inflammation and oxidative stress. Sci Total Environ. 2023;869:161760.
68. Ma J, Zhang J, Zhang Y, Wang Z. Causal effects of noise and air pollution on multiple diseases highlight the dual role of inflammatory factors in ambient exposures. Sci Total Environ. 2024;951:175743.
69. [http://www.stats.gov.cn/sj/ndsj/]
70. Labor, Markets. [https://www.worldbank.org/en/topic/labormarkets]
71. Morris NB, Jay O, Flouris AD, Casanueva A, Gao C, Foster J, Havenith G, Nybo L. Sustainable solutions to mitigate occupational heat strain - an umbrella review of physiological effects and global health perspectives. Environ Health. 2020;19(1):95.
72. Achebak H, Rey G, Lloyd SJ, Quijal-Zamorano M, Fernando Méndez-Turrubiates R, Ballester J. Drivers of the time-varying heat-cold-mortality association in spain: A longitudinal observational study. Environ Int. 2023;182:108284.
73. Sinha P, Coville RC, Hirabayashi S, Lim B, Endreny TA, Nowak DJ. Variation in estimates of heat-related mortality reduction due to tree cover in U.S. cities. J Environ Manage. 2022;301:113751.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.